Backtesting in Quantitative Investing

Politecnico di Milano

Mathematical Engineering – Financial Engineering, 2025/2026

01

Backtesting Best Practices

Data hygiene, research process, and strategy development guidelines

Data Quality & Sourcing

High-quality data is essential to backtesting. The search for better data is a never-ending task. When comparing vendors, ask the following questions:

Definition & Interpretation

What does the data mean? What are the physical units, reference currency, and time unit? Mistakes from unit conversion errors are common.

Provenance

Where is the data coming from? Does the vendor collect it directly (e.g., web scraping) or act as an intermediary? Is sampling exhaustive?

Completeness

Are there obviously or non-obviously missing data points? For example, intermittently missing prices or unrecorded transactions.

Quality Assurance

Does the provider have checks for data quality consistency? Are there change points in the data characteristics over time?

Point-in-Time vs. Restated

Does the provider offer data as of the original date, without later corrections? Restated data is a source of data leakage.

Transformations

How is data processed? Imputation of missing values, winsorization, outlier removal, price calculations (mid bid-ask, VWAP).

Always explore alternatives and complements: can you get larger coverage, deeper history, or complementary datasets?

Avoiding Data Leakage

Data leakage is the presence in the training data of information available up to time t that is contained in the target (returns in t+1 and later). The reference rule: never use data in a backtest that you could not have used in production on that date.

1. **Survivorship Bias**
   Backtesting only on stocks that survived to the end of the period biases results upward. Survivors tend to have higher liquidity, momentum, and size. Remedy: use an investment universe applicable at each point in time.

2. **Financial Statements**
   Include financial data only on the day of public release, not the quarter it refers to. Reporting lags vary and using future-dated data inflates backtest returns.

3. **Point-in-Time Data**
   Always use the most recent data available as of that date. If a 10-Q is restated later, the backtest should not reflect that — test robustness to errors in input data.

4. **Price Adjustments**
   Splits and dividends require adjusted prices for return computation but raw prices for feature generation. A low historical price after a split does not indicate a cheap stock.

5. **Missingness**
   Missing data points may be look-ahead biased (withheld and later released) or informative of future returns. Investigate the cause before imputing or dropping.

Strategy Development Guidelines

Qualitative recommendations that, while not formally provable, are grounded in decades of practitioner experience.

Have a Theory

Pre-register predictions before running the backtest. A theoretical foundation reduces the search space, makes results interpretable, and allows critique and revision.

Enforce Reproducibility

Document all strategies completely so they can be reproduced and rerun at any time. Code, parameters, data versions — everything should be versioned.

Match Backtest to Production

Use the same point-in-time data, optimization formulations, market impact model, and codebase in backtesting and in live trading.

Calibrate the Market Impact Model

The market impact model has a descriptive role in backtests. Calibrate its parameters against live performance so that realized and backtested PnL overlap.

Include Borrow Costs

Borrow costs for short positions can materially affect the short side's profitability. Historical borrow rates are hard to obtain; approximate or predict them.

Define the Protocol Beforehand

The backtesting protocol should be changed rarely, for good reason, and when it changes, all strategies should be re-evaluated under the new protocol.

02

Common Pitfalls in Backtesting

Cross-validation, walk-forward, and the multiple testing problem

Cross-Validation vs. Walk-Forward

Two common evaluation schemes for trading strategies. Neither is without flaws.

Cross-Validation

How it works:

Split training data into K folds. Train on K−1 folds, test on the remaining. Repeat and average performance.

Advantages:

- Uses all available data for both training and testing
- Good for large parameter spaces

Drawbacks:

- Samples are not independent (serial dependence in returns)
- Temporal folds can leak target information (e.g. momentum signals leak past returns into the training set)
- Holdout sample may become “just another variable to optimize”

Walk-Forward

How it works:

Train on data up to time t, test on period t+1. Roll the window forward and repeat. Mirrors the production process.

Advantages:

- Non-anticipative: immune from temporal data leakage
- Respects serial dependence in returns
- Naturally adaptive to changing market regimes

Drawbacks:

- Uses less training data than cross-validation
- Delay in running the strategy in production (opportunity cost)
- Severe limitation when model/parameter space is large

When enough strategies are tested, some will appear profitable by chance. This is the fundamental challenge of backtesting.

03

The Role of Transaction Costs

Broker fees, bid-ask spreads, slippage, and market impact can erode or eliminate backtest alpha

Anatomy of Transaction Costs

Transaction costs can be similar in magnitude, or even outweigh, the systematic premia a strategy aims to capture.

Broker Commissions & Fees

Fixed or %

Fees levied by brokers for executing trades. Can be a fixed fee per trade, a percentage of trade value, or per-share. Includes exchange fees, clearing costs, and regulatory charges. Most predictable and easiest to model.

Bid-Ask Spread

1–50+ bps

The difference between the best bid and ask price. Represents the cost of immediacy. Wider for illiquid names, small caps, and during volatility spikes. A natural floor on round-trip trading cost.

Slippage

Variable

The difference between the intended execution price and the actual fill price. Depends on order type, market conditions, latency, and execution algorithm. Can be substantial for market orders in volatile conditions.

Market Impact

~√Volume

The price movement caused by your own order. Larger orders move the market more. Grows approximately as the square root of order size relative to daily volume. The hardest cost to model and the most dangerous to ignore.

Modeling Costs: From Backtest to Production

Cost Modeling Approaches

Flat Cost Model

Fixed cost per trade (e.g. $0.005/share). Simple and predictable, but ignores order size effects and market conditions.

Linear Model

Cost proportional to trade value (e.g. 5–10 bps). Captures the bid-ask spread component well but underestimates large-order impact.

Square-Root Model

Impact $\propto \sigma \cdot \sqrt{Q / V}$, where σ is volatility, Q is order size, and V is daily volume. Industry-standard for market impact modeling.

Calibrated / Vendor Model

Third-party models (e.g. from broker or vendor) fitted to real execution data. Must be calibrated against your own live performance — never take vendor estimates at face value.

Alignment Principles

Backtest → Production Gap

The largest source of live trading disappointment is the gap between backtested and realized transaction costs. Closing this gap is critical.

Calibrate Against Live PnL

Adjust market impact model parameters so that realized and backtested PnL overlap as closely as possible for the current strategy version.

Cost-Aware Optimization

Portfolio construction should balance expected alpha against estimated transaction costs. High-turnover strategies with thin alpha are especially vulnerable.

Borrow Costs on the Short Side

Short positions incur borrow fees that can materially erode short-side profits. Hard-to-borrow names can cost 5–20%+ annualized. Include realistic estimates.

Key Takeaways

1. Data quality is the foundation of every backtest. Understand provenance, completeness, and transformations before using any dataset.
2. Data leakage is the most impactful pitfall — survivorship bias, look-ahead bias, and point-in-time errors can make worthless strategies look profitable.
3. Neither cross-validation nor walk-forward backtesting is perfect. Use them as complements.
4. Multiple testing inflates false discovery rates
5. Transaction costs (broker fees, bid-ask spread, slippage, and market impact) can match or exceed the alpha a strategy captures. Ignoring them is the fastest way to turn a profitable backtest into a losing live strategy.
6. Calibrate your cost model against live trading performance, and always define the full backtesting protocol — including the cost model — before running any tests.

Statistical Arbitrage in the EMU Equity Market

Lab Introduction & Toolkit

Politecnico di Milano — Financial Engineering
FY 2025/2026

01

PCA-based Factor Models

Extracting systematic risk factors from the correlation matrix

You will decompose stock returns into systematic and idiosyncratic components using Principal Component Analysis on the sample correlation matrix of Euro Stoxx 50 returns.

Multi-Factor Decomposition

Separating systematic movements from stock-specific behavior

The Factor Model

$R_i = \alpha_i + \Sigma \beta_{ij} F_j + \tilde{R}_i$

Each stock return is decomposed into:

- A drift term α_i (idiosyncratic excess return)
- Systematic exposure to m common factors
- An idiosyncratic residual $\tilde{R}_i$

The factors F_j capture market-wide risk.

Why PCA?

Data-driven factor extraction

PCA extracts factors from the correlation matrix without requiring prior economic knowledge. Eigenvalues identify significant vs. noise factors. Eigenportfolios can be interpreted as long-short positions at the industry/sector level.

Standardized Returns & Correlation Matrix

Returns are standardized by stock-level mean and std before PCA. We use a 252-day (1-year) rolling window. The first m eigenvalues are "significant"; the rest form the noise spectrum.

Key insight: PCA factors are unbiased by market cap — unlike ETF-based factors, they give no a priori weight to large-cap stocks.

Eigenportfolios & Factor Selection

Interpreting the principal components and choosing how many factors to retain

1 First Eigenportfolio

Proxy for the market portfolio

All coefficients positive, weighted inversely to volatility. Behaves like a capitalization-weighted index. Explains the largest share of total variance.

2 Higher Eigenportfolios

Sector-level long-short portfolios

Components have mixed signs and exhibit "coherence": nearby coefficients belong to the same industry. Interpretable as sector rotations.

3 Fixed Factor Count

Use m = 4 principal components

The assignment uses the first 4 PCs as factors. These capture the dominant systematic movements while leaving the idiosyncratic component for trading.

4 Variable Factor Count

Target a fraction of explained variance

Sensitivity analysis: vary the number of PCs to target 40%, 55%, 65%, or 75% of explained variance. Variance explained by a fixed m fluctuates over time.

02

The Ornstein-Uhlenbeck Process

Modeling mean-reversion in the idiosyncratic residuals

The cumulative residuals from the factor model are modeled as a continuous-time mean-reverting process, whose parameters are estimated via a discrete AR(1) regression on a 60-day window.

Ornstein-Uhlenbeck Dynamics

A continuous-time model for mean-reverting residuals

The SDE

$dX_i(t) = \kappa_i(m_i - X_i(t))\, dt + \sigma_i\, dW_i(t)$

Parameters (stock-specific):

- κ_i : mean-reversion speed (higher = faster)
- m_i : equilibrium level
- σ_i : diffusion volatility

Equilibrium distribution: Normal with

$E[X_i] = m_i$ and $Var[X_i] = \sigma^2_i / (2\kappa_i)$

Half-life: $\ln(2) / \kappa_i$

Discrete Estimation via AR(1)

$X(t+\Delta t) = a + b \cdot X(t) + \varepsilon$

Regress $X(t+1)$ on $X(t)$ over 60-day window.

Map regression coefficients to O-U params:

- $\kappa = -\ln(b) / \Delta t$ (mean-reversion speed)
- $m = a / (1 - b)$ (equilibrium level)
- $\sigma^2_{eq} = Var(\varepsilon) / (1 - b^2)$

Require $\kappa > 252/30 = 8.4$ to ensure fast mean-reversion.

Cumulative Residuals as Input

The O-U model is fitted on the cumulative sum of daily residuals from the factor model, not the raw daily residuals themselves. This gives a process $X_i(t)$ that oscillates around m_i.

Key insight: The O-U framework provides a principled way to distinguish mean-reverting stocks from trending ones via κ.

03

S-Score & Trading Signals

From residual dynamics to actionable trading rules

The s-score normalizes the current residual by its equilibrium mean and volatility, producing a dimensionless signal. A modified version incorporates the drift for a built-in momentum overlay.

The S-Score

A dimensionless measure of dislocation from equilibrium

Pure Mean-Reversion S-Score

$s_i = (X_i(t) - m_i) / \sigma_{eq,i}$

Measures how many equilibrium standard deviations the residual is from its mean.

$\sigma_{eq,i} = \sigma_i / \sqrt{2\kappa_i}$ is the equilibrium standard deviation of the O-U process.
Dimensionless → comparable across stocks.

Modified S-Score (with Drift)

$s_{mod,i} = s_i - \alpha_i / (\kappa_i \cdot \sigma_{eq,i})$

Incorporates the residual drift α_i.
A positive drift makes it harder to short ($s_{mod} < s$) and easier to go long.

Acts as a built-in momentum filter: stocks with strong drift require larger dislocations.

Trading Rules

Threshold-based entry and exit for long and short positions

Open long if $s_{mod} < -1.25$ | Open short if $s_{mod} > +1.25$ | Close long if $s_{mod} > -0.50$ | Close short if $s_{mod} < +0.50$

Positions are opened/closed at the close of the day following the signal. Equal weighting across active signals, 100% gross exposure.
Sensitivity: test alternative opening thresholds of 1.0, 1.5, and 2.0.

04

Backtesting & Extensions

Rolling estimation, transaction costs, and trading-time adjustment

You will implement a daily rolling backtest with walk-forward estimation, compare performance with and without transaction costs, and explore volume-adjusted returns.

Rolling Backtest Framework

Daily re-estimation with walk-forward design

1 Daily Re-Estimation

Factor model + O-U at every trading day

At each date: re-estimate PCA on a 252-day window, compute residuals, fit O-U on the last 60 days, update s-scores. Positions carry state across days.

2 Position Tracking

Maintaining state across rebalance dates

Positions persist until an explicit close signal fires. This makes the strategy path-dependent: the backtest must track each stock's open/close status sequentially.

3 Transaction Costs

5 bps all-in per trade

Compare zero-cost and 5 bps scenarios. Costs are proportional to absolute weight changes. High turnover strategies suffer most from this friction.

4 Portfolio Construction

Equal-weight with 100% gross exposure

Active signals get equal weight, scaled to target 100% gross exposure. No leverage optimization or sector tilting — keep it simple.

Sensitivity Analysis & Trading Time

Robustness checks and the volume-adjustment extension

Entry Threshold Sensitivity

How robust is the strategy to the s-score cutoff?

Test alternative opening thresholds:

- $s_{bo} = 1.0$ (more aggressive entry)
- $s_{bo} = 1.5$ (more conservative entry)
- $s_{bo} = 2.0$ (highly selective entry)

Trade-off: lower threshold → more trades but weaker signal; higher → fewer but stronger.

Variable Number of Factors

Targeting explained variance instead of fixed m

The variance captured by 4 PCs varies over time.

Alternative: choose m to meet a target level:

- 40% (few factors, large residual)
- 55% (moderate extraction)
- 65% (balanced split)
- 75% (heavy systematic removal)

Trading Time: Volume-Adjusted Returns (Optional)

$\bar{R}_{i,t} = R_{i,t} \cdot \bar{V}_i / V_{i,t}$ — weighting returns by inverse relative volume

The paper proposes measuring returns in "trading time": price moves on low-volume days carry more information.

This accentuates contrarian signals on quiet days and dampens them on high-volume days.

Key insight: Volume adjustment acts as an implicit confidence filter — trust price dislocations more when fewer participants are trading.

Deliverables & Practical Notes

1. **Deliver by Friday 17 April, 18:00**
   to financial.engineering.polimi@gmail.com with subject "BS: Assignment 4a, Group #".

2. **Submit a short PDF with results**
   and a discussion of the strategy performance and sensitivity analysis.

3. **Submit the Python code**
   using your preferred IDE. Comment in detail and use descriptive variable names.

4. **Use Euro Stoxx 50 data provided**
   sx5e_underlyings.csv, volume.csv, ticker_details.csv. Assume total return indices in EUR, simple returns, ignore risk-free rate.

5. **Fill the gaps and correct errors**
   in case of incoherences between the assignment PDF and the code, follow the assignment document.

Sources

1. **Statistical Arbitrage in the U.S. Equities Market**
   Marco Avellaneda & Jeong-Hyun Lee (2008)
   The reference paper: PCA approach, O-U process, s-score, trading-time signals

2. **The Elements of Quantitative Investing**
   Giuseppe Paleologo
   Backtesting best practices and transaction cost modeling

3. **Advances in Financial Machine Learning**
   Marcos López de Prado
   Multiple testing, walk-forward validation, and data leakage pitfalls